Effect of Plant Roots and Root Channels on Solute Transport

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ABSTRACT

SINCE many of the practical applications of solute modeling are to situations where crop roots are present, it is necessary to investigate the effect of roots and void root channels on the transport of solutes through soils. In an attempt to do this chloride breakthrough experiments were conducted over a two year period in soil columns packed uniformly with sandy loam or loamy sand. A series of experiments were conducted in bare soil, followed by repeated experiments with wheat plants in the columns, and finally in bare soil again after the wheat crop had died. The loamy sand columns showed significantly higher dispersion than the sandy loam columns in the precrop experiments, but the effective dispersion decreased when the experiments were repeated with plant roots present. The post crop experiment showed the highest dispersion of any of the studies possibly due to the presence of old root channels in the soil. Very little immobile water was present in the precrop experiments, but both the cropped experiments and the post crop experiments exhibited significant pore bypass; immobile water fractions ranged from 11 percent to 39 percent of the total water content.

INTRODUCTION

As a result of crop fertilization, waste disposal operations and other land based practices, large quantities of soluble chemicals are applied each year to soil surfaces. Some of the soluble materials may have high affinities for the soil matrix, while others move relatively freely with the soil solution. To estimate the hazard associated with nutrient movement below crop roots or chemical percolation below waste disposal sites, it is essential to understand the physical processes controlling the transport of these substances through the soil to underlying ground waters. To this end, a number of studies have been conducted looking at solute transport through bare soils, both under uniform conditions (Biggar and Nielsen, 1967) and in artifically or naturally aggregated porous media (Rao et al., 1980). However, very little is known about the influence of plant roots on solute movement in soils.

Although our theoretical and experimental knowledge has been largely confined to bare soils, some of the useful concepts that have evolved may be applicable to conditions where a crop is present. In uniform soils, solute transport may be adequately described using a simple convection-dispersion model (Biggar and Nielsen, 1967); however, if the soils are structured or contain flow channels one must use a theoretical representation which includes the effect of pore bypass or immobile water (the water adjacent to soil grains or held in dead-end pores), (van Genuchten and Wierenga, 1976; Gaudet et al., 1977; El Prince and Day, 1977). To represent a media whose structure has been altered by roots, one must alter the basic convection-dispersion equation. To accomplish this, several parameters and their interrelationships need to be evaluated: the average pore water velocity V, the dispersion coefficeint D, and the rate of salt diffusion into or out of stagnant or immobile regions. Under uniform pore size conditions, the average pore water velocity (V) will be equal to the ratio of the water flux (J_w) and the volumetric water content (θ_v): $V = J_w/\theta_v$. When substantial pore bypass occurs due to the presence of aggregates or natural channels, the average velocity will exceed this ratio because only part of the wetted pore space is utilized for transport. This situation is often represented in models by assuming that the total water content θ_v is equal to the sum of the mobile (θ_m) and immobile (θ_{im}) water content; as a consequence, the average pore velocity is set equal to J_w/θ_m. However, the mobile water content cannot be measured directly but must be inferred from the breakthrough time of the solutes. In addition, the dispersion coefficeint may be represented as being proportional to the water velocity ($D=\lambda V$), when molecular diffusion in negligible. This proportionality factor is referred to as the dispersivity and may be constant for a particular soil (Bresler and Laufer, 1974). However, the validity of this latter assumption is dependent upon both the frequency and magnitude of the water inputs. For most soils receiving frequent irrigations the water flux will be large and molecular diffusion may be negligible. As a consequence, it may be possible to detect changes in the flow geometry caused by plant roots by observing changes in the dispersivity λ.

When a plant is introduced into a porous medium, the most obvious effect is to concentrate the soil solution by preferentially removing water and leaving solutes behind. Most models that attempt to simulate solute movement through a crop root zone use the basic convection-dispersion equation, altered to allow for water uptake. Little or no work has been conducted on the influence of plant roots on the physical structure of the porous medium or on the subsequent effect of void root channels on water movement and solute transport (Jury et al., 1976; Childs and Hanks, 1975).

The purpose of this study was to examine the influence of plant roots on the solute transport parameters described above. This was accomplished by conducting a series of chloride breakthrough experiments in identical soil columns, first, under preplant conditions, second,

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TABLE 1. TEXTURAL ANALYSIS AND PHYSICAL CHARACTERISTICS OF SOILS USED IN THE EXPERIMENTS.

Soil type	% sand	% silt	% clay	Porosity	Bulk density, g/cm^3
Sandy loam	53.5	36.3	10.2	0.44	1.52
Loamy sand	70.0	27.0	3.0	0.44	1.52

with living plants present, and third, after the plant roots had died and decayed.

MATERIALS AND METHODS

Experimental studies were conducted in six soil colums using a Pachappa sandy loam and a Pachappa loamy sand (fine-loamy, mixed, thermic, mollic hapluxeralf). Three of the columns were 60 cm in length with an internal diameter of 13.5 cm. The remaining three columns were 102 cm in length with an internal diameter of 15.3 cm. Although the columns varied in length, it is likely that both sets of columns were sufficiently large so as to minimize any apparatus induced error (James and Rubin, 1972). Each column was uniformly packed to a bulk density of 1.52 g/cm^3. The procedure used to pack the columns was checked by gamma ray attenuation and is described in Gish, 1981. The textural analysis and physical characteristics of the materials used are given in Table 1.

For the bare soil experiments conducted prior to growing plants, a 0.015N calcium sulphate solution was used as the irrigation water and was applied uniformly to the soil surface through an irrigation network made up of six stainless steel tubes. The three loamy sand column experiments were conducted at water flux densities of 1.5, 3.9, and 5.3 cm/day in 1978, while the sandy loam experiments were conducted at 0.5, 1.0, and 4.5 cm/day in 1979. For both studies, when steady state conditions were established, a 5 meq/L chloride solution was added to the inlet solution, at the same density as the initial solution. Drainage water was removed from the bottom of the column at a modest suction sufficient only to insure unsaturated conditions. When the chloride concentration in the effluent was identical to the concentration of the irrigation water, the experiments were terminated and soil cores were taken within the column and analyzed for volumetric water content. The chloride concentration in the effluent was used to determine the transport coefficients. The remaining soil was removed from the columns, air dried, and sieved (through a 2 mm screen). Next, the columns were then repacked to the same bulk density with a preplant fertilizer mixed ino the top 7 cm.

Subsequent to the bare soil experiments, Anza, a salt tolerant, semi-dwarf wheat (Triticum aestivum) was planted in each of the six columns. Irrigation of the plants proceeded as in the bare soil experiments except that a 0.1 strength Hoagland solution was added to avoid any nutrient deficiencies. These experiments were conducted in a greenhouse where solar radiation, air temperature, and relative humidity were continuously recorded in order to estimate potential evapotranspiration by the modified Penman equation (Doorenbos and Pruitt, 1975). This estimate of evapotranspiration, modified for the ratio of lateral leaf area to soil column area, was used to guide irrigation management so as to keep the leaching fractions (LF) constant at 0.15, 0.2 and 0.3.

After a constant leaching fraction had been established for two weeks, a bromide solution was added to the irrigation (smaller amounts of calcium sulfate were added so as to maintain the same solution density).

The bromide concentration in the soil solution was used to determine the water uptake distribution as a function of depth (Gish and Jury, 1982). After the bromide output flux equaled the bromide input flux, the bromide soil solution concentration was determined by extracting 20 ml of soil solution from four depths in the columns (7, 35, 64, and 91 cm). However, the two columns subjected to the 0.15 leaching fraction failed to reach steady state, and, as a consequence, were taken down and dismantled. Subsequent to the determination of the water uptake distribution, a 5 meq/L chloride solution was introduced into the irrigation water of the remaining four columns. The effluent was collected on a daily basis and analyzed for chloride. The chloride and bromide concentrations were distinguished by determining the total halide concentration and subtracting the bromide concentration which was determined with a bromide specific ion electrode.

After the soil plant experiment was terminated, two of the sandy loam columns were placed intact, in storage for 1.5 years, after which the bare soil chloride displacement studies were repeated to determine the influence of root channels on solute movement. The two remaining loamy sand columns were dismantled and analyzed for the average water content.

THEORY

The solute transport model used in our crop experiments in an extension of the mobile-immobile water model of Van Genuchten and Wierenga (1976). The starting point for such a model is the mass balance for water and solute equations [1] and [2]

$$\frac{\partial \theta m}{\partial t} + \frac{\partial \theta_{im}}{\partial t} + \frac{\partial J_w(z)}{\partial z} = -Sw(z) \quad [1]$$

$$\frac{\partial(\theta_m C_m)}{\partial t} + \frac{\partial(\theta_{im} C_{im})}{\partial t} + \frac{\partial J_s(z)}{\partial z} = 0 \quad [2]$$

where C_m is the concentration of solute in the mobile phase, C_{im} is the concentration of solute in the immobile phase, $J_w(z)$ is the water flux, $J_s(z)$ is the solute flux, and $S_w(z)$ is the water uptake rate per unit soil volume. Equation [2] assumes that essentially no solute is taken up within the porous medium. We assume also that the immobile water content is constant. During steady state water flow ($\partial \theta_m / \partial t = 0$), which is a reasonable approximation under high frequency irrigation when root growth has essentially ceased, the water flux at any depth will be a function of the application rate (I) and the water uptake distribution $S_w(z)$:

$$J_w(z) = I - \int_o^z S_w(z) dz \quad [3]$$

The water uptake distribution as a function of depth in the columns was measured by analyzing the steady state bromide concentrations as described in Gish and Jury, 1982. Essentially, this model assumes that the plant is mature and receives frequent irrigations. It is mathematically represented in equation [4] as an exponentially decreasing function with depth.

$$S_W(z) = \frac{ETB}{(1-R)L}(\exp - BA) \qquad A = 0, \frac{z}{L}, \frac{2z}{L}, \frac{3z}{L}, \frac{4z}{L} \ldots 1 \quad . \quad [4]$$

where R is the water uptake shape parameter $S_w(L)/S_w(O)$, ET is the evapotranspiration rate, $B = -\ln(R)$, and A is the dimensionless depth representing a depth z.

The vertical solute flux $J_s(z)$, assumed to occur entirely in the mobile phase, is given by equation [5].

$$J_{s(z)} = -\theta_m D_{(z)} \frac{\partial C_m}{\partial z} + J_{w(z)} C_m \ldots\ldots\ldots\ldots\ldots [5]$$

where the dispersion coefficient $D_{[z]}$ is represented for simplicity as a linear function of the water velocity $V_{[z]}$

$$D_{(z)} = D_o + \lambda V_{(z)} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots [6]$$

where D_o is the molecular diffustion coefficient. At water velocities commonly found in soils, D_o is usually negligible compared to the second term (Dagan and Bresler, 1979). As a consequence, we modeled dispersion as proportional to the water velocity, $(D_{[z]} = \lambda V_{[z]})$.

Since equation [2] contains two unknowns C_m and C_{im}, an additional equation describing changes of solute in the immobile phase is needed to fully specify solute transport. The model used is the diffusional mass transfer equation proposed by Coats and Smith (1964).

$$\frac{\partial \theta_{im} C_{im}}{\partial t} = \alpha(C_m - C_{im}) \ldots\ldots\ldots\ldots\ldots\ldots\ldots [7]$$

where α is the diffusional mass transfer coefficient which describes the rate of exchange of solute between the two phases.

The system of equations [1] to [8] were solved subject to the following boundary conditions:

$$C_m(0,t) = C_o \qquad \text{upper boundary} \ldots\ldots [8a]$$

$$\frac{\partial C_m}{\partial z}_{(L,t)} = \frac{\partial C_{im}}{\partial z}_{(L,t)} = 0 \qquad \text{bottom boundary} \ldots\ldots [8b]$$

$$C_m(z,o) = C_{im}(z,o) = 0 \qquad \text{initial condition} \ldots\ldots [8c]$$

where L is the column length.

The system of equations [1] through [8] were used in all experiments to fit observed chloride breakthrough to predicted values by adjusting λ, α, and θ_m until the standard error of estimate was minimized. In the bare soil experiments $S_w(z)$ was set equal to 0. A Crank-Nicolson finite difference procedure was used to solve the model. The solution to the equation was tested for accuracy by setting α = o and comparing the numerical results with results based on the analytic solution given by El Prince and Day (1977). In the comparison, these two models gave identical results. For $\alpha \neq$ o, less than 0.01 percent error in mass balance was observed in the calculations. On the basis of these comparison it was assumed that the numerical solution to equation [1] through [8] was accurate.

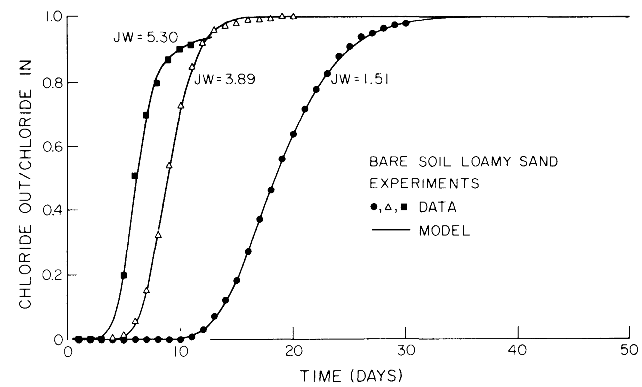

Fig. 1—Chloride breakthrough curves in precrop loamy sand experiments.

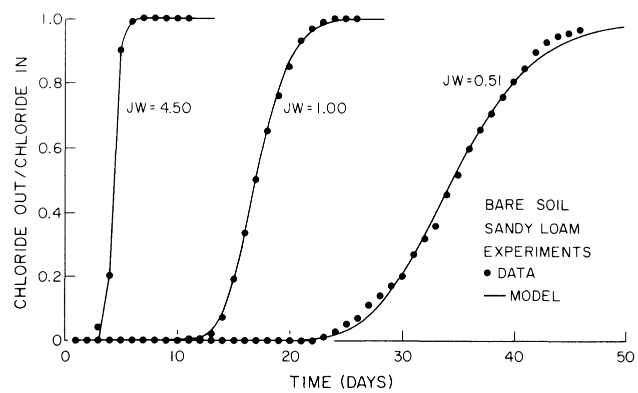

Fig. 2—Chloride breakthrough curves in precrop sandy loam experiments.

RESULTS

Figs. 1 and 2 show the measured and calculated breakthrough curves for the bare soil experiments conducted before the plant was grown in the columns. Each of the experimental input and soil conditions, as well as the transport parameters, are summarized in Table 2. In only one of the six bare soil precrop experiments, that of the highest water flux (6.3 cm/day), was there any evidence of mass transfer between the mobile and immobile water present. Each of the loamy sand experiments showed breakthrough curves characteristic of a high dispersivity, between 2.5 and 2.7 cm, whereas the dispersivities for the sandy loam study ranged between 0.5 and 0.9 cm. These values are well within the range of values previously reported in the literature (Biggar and Nielsen, 1976; De Smedt and Wierenga, 1978). In addition, all of the precrop bare soil experiments exhibited mobile water contents greater than 91% of the total volumetric water content with the exception of the high water flux study in the loamy sand.

Fig. 3 shows the chloride breakthrough curves for the two cropped loamy sand columns. As would be expected, the 0.3 leaching fraction experiment exhibited a shorter transit time. No mass transfer effects were needed to describe the breakthrough curves, but the dispersivities dropped substantially from their values under the precrop conditions. In addition, there was evidence of increased immobile water when the crop was growing, a phenomena discussed in some detail by Gish and Jury (1982).

Fig. 4 shows the breakthrough curve for one the sandy loam columns taken after the plant roots had decayed,

TABLE 2. EXPERIMENTAL CONDITIONS AND MEASURED TRANSPORT PARAMETERS FOR CHLORIDE BREAKTHROUGH EXPERIMENTS.

Experiment	Soil	J_w cm/day	θ_v cm^3/cm^3	θ_m/θ_v %	λ cm	α 1/day	V cm/day	D cm^2/day
Precrop	loamy sand	1.51	0.30	0.94	2.7	~0*	5.4	14.2
"	" "	3.89	0.36	0.97	2.5	~0	11.2	27.6
"	" "	5.30	0.38	0.85	2.5	0.01	15.7	38.6
"	sandy loam	0.51	0.31	0.97	0.9	~0	1.7	1.5
"	" "	1.00	0.32	0.91	0.6	~0	3.5	2.0
"	" "	4.50	0.35	0.97	0.3–0.5	~0	13.5	4.2– 6.8
Crop LF=0.2	loamy sand	3.75	0.25	0.83	0.5	~0	3.6–18.2	1.8– 9.1
Crop LF=0.3	" "	4.40	0.26	0.87	0.6	~0	5.9–19.6	3.5–11.8
Post crop	sandy loam	2.50	0.33	0.89	3.6	~0	8.5	31.0
Post crop	" "	4.60	0.33	0.61	2.8	~0	20.6	58.0

*α<0.0001/day

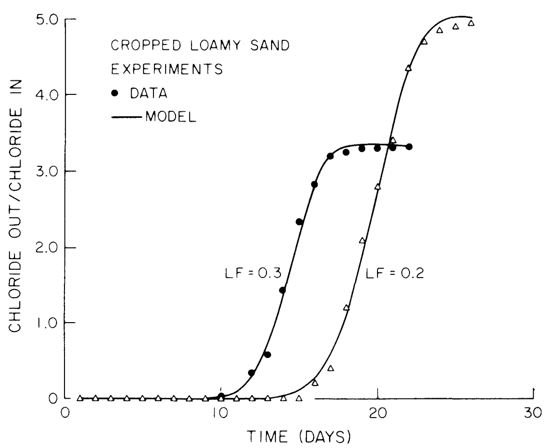

Fig. 3—Chloride breakthrough curves in loamy sandy cropped experiments.

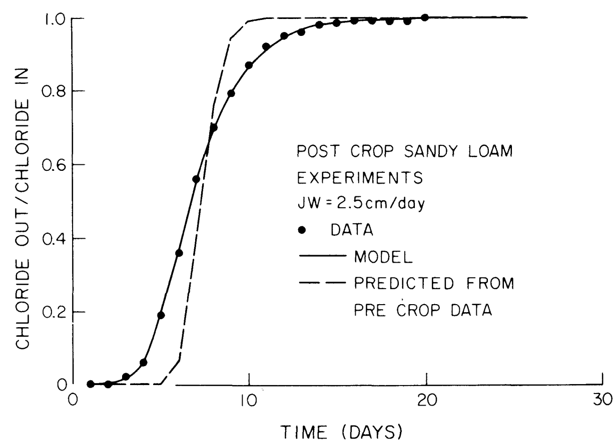

Fig. 4—Chloride breakthrough curves in postcrop sandy loam experiment for J_w = 2.5 cm/day. Dashed curves uses coefficients measured in precrop experiments to predict breakthrough.

leaving behind void root channels. Also shown in this figure is the predicted breakthrough curve for the input water flux which would have occurred had the properties of the porous medium not changed from the values and trends previously established in the three precrop bare soil experiments. Fig. 5 shows the corresponding breakthrough curve for another sandy loam study conducted at a water flux of 4.6 cm/day. Essentially, no mass transfer effects were required to explain these breakthrough curves, but the dispersivities in both columns increased substantially over their values in the base soil experiment before cropping. In addition, the higher water flux (4.6 cm/day) exhibited substantial effects of pore bypass, dropping the mobile water fraction down to 61 percent.

DISCUSSION

All of the solute breakthrough curves showed excellent agreement with the numerical model based on equations [1] to [8] with $\alpha = 0$. As a consequence, a two parameter (i.e., V. and D) model with an analytical solution for bare soil such as the one proposed by El Prince and Day (1977) would accurately describe most of the bare soil breakthrough experiments reported here. This is to be expected as the columns were uniformly packed, unaggregated, and the displacing solutions used were of equal density. In addition, the trends between the columns with the same texture suggests that the experimental procedure eradicated any apparatus induced error. Thus, we conclude that the process of repacking did not affect the transport coefficients.

Although the model used here to simulate the precrop experiment was similar to the one proposed by Gaudet et al. (1977), we obtained a completely different set of results than did those authors. For example, they consistently observed marked asymmetry in their chloride breakthrough curves, and required mass transfer coefficients from .7 to 2.4/day to fit the model to the data. We had only one of the bare soil experiments with a mass transfer coefficient above 0.01/day. These authors, however, conducted their experiment in pure sand at extremely high water flux densities (J_w = 47 to 259 cm/day) and also had a significant density difference between the initial and displacing solutions. This latter feature has been shown to create instabilities which result in asymmetric breakthrough curves (Krupp and Elrick, 1968). Since we took precautions to insure that

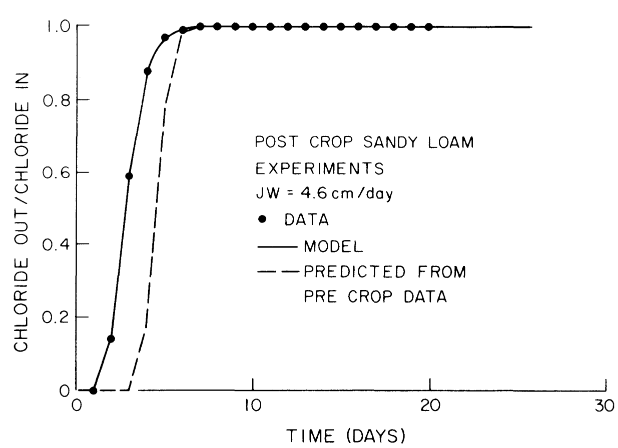

Fig. 5—Chloride breakthrough curves in postcrop sandy loam experiment for J_w = 4.6 cm/day. Dashed curves used coefficients measured in precrop experiments to predict breakthrough.

no density differences existed, we saw little or no asymmetry in our breakthough curves. Gaudet et al. (1977) also observed much lower mobile water fractions in their flow experiments than observed in this study. However, this too may be a by-product of high velocities and large density differences in the displacing solutions.

The dispersivity was less in the sandy loam columns (0.5 to 0.9 cm) than in the loamy sand (2.5 to 2.7 cm) experiments which were conducted at comparable flow rates. This suggests a greater range of local pore water velocities in the coarser-textured soil, and that some of the larger pores are carrying enough solution to cause the early breakthrough edge characteristic of displacement experiments with high dispersion or solute bypass in the macropores. In addition, the dispersivity λ, appears to remain relatively constant regardless of the water flux density. This suggests that for a homogeneous medium, the dispersivity may be a function of soil texture.

When the loamy sand columns were used in the crop experiments, the dispersivity coefficient decreased from an average of 2.56 cm for the bare soil experiment to 0.55 cm, even though the range of average pore water velocities within the root zone was similar to those imposed on the bare soil experiments. A possible explanation for the abrupt drop in the dispersivity is that the plant roots effectively homogenized the porous medium by excluding flow through some of the large pores. Thus, the presence of living plant roots seem to have decreased dispersion during transport through the root zone, which is contrary to earlier speculations that the roots would increase dispersion.

When the sandy loam columns, which had been highly uniform and well behaved during the precrop experiments, were retested after the wheat had died the effective dispersion increased significantly. The dispersivities of 2.8 and 3.6 cm are characteristics of a porous medium with significant flow occurring in large pores, such as the loamy sand experiment prior to cropping. In addition, there is evidence of a substantial amount of pore bypass ($\theta_m/\theta_v = 0.6$ to 0.9) indicating that the bulk of the water flow was taking place in a few channels. However, even with substantial bypass, no mass transfer effects were required to explain the observed breakthrough curves.

The displacement experiments were initially conducted under relatively isotropic homogeneous conditions in order to isolate the effects of the plant roots on solute transport. The two soils were from the same soil series and general location but differed only in the sand, silt, and clay fractions present (Table 1). These differences, though not dramatic, seem to have created large differences in the dispersion coefficients between the two soils. The introduction of plant roots into the soil seemed to have either closed or occupied the larger pores of the soil resulting in a more uniform network of flow channels. Though this latter process seems rather speculative, it was also noted by Nye and Tinker, 1977. This in turn may have been responsible for a smaller dispersion coefficient but a greater fraction of immobile water. Channels created by the decayed plant roots tended to form macopores through which the solutes could flow preferentially at high water flux densities. Thus, in any field situation, it is likely that the large range of vacant root channels and active roots present would counteract each other. In fact, as reported in Gish and Jury (1982), plant roots do seem to increase the immobile water fraction under some natural field conditions as well as in our laboratory experiments.

CONCLUSION

This study demonstrates that solute movement in the presence of living or dead plant roots can differ significantly from that under uniform conditions, showing in particular, that the fraction of immobile water and the hydro-dynamic dispersion of the porous medium change substantially. Living plant roots seem to either fill the larger pores or cause them to decrease in size by rearranging the soil grains during the growth process. This effect could lower the dispersion coefficient and create a narrower range of pore water velocities than would be present in the same soil prior to cropping. After the roots decayed, the macropores formed by the roots may be most important in solute transport at large water flux densities, thus increasing dispersion and decreasing the transit time to underlying ground waters.

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NOTATION

α mass diffusional transfer coefficient, day^{-1}

(continued on page 451)

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(continued from page 444)

a — determined constant, cm^3 solution/cm^3 soil, day).
b — determined constant, cm^{-1}
B — function of the water uptake shape parameter, dimensionless.
C — solute concentration, meq salt/cm^3 solution.
C_{im} — solute concentration in the immobile phase, meq salt/cm^3 solution.
C_m — solute concentration in the mobile phase, meq salt/cm^3 solution.
D_o — molecular diffusion coefficient, cm^2/day.
D — effective dispersion coefficient, cm^2/day.
ET — evapotranspiration rate, cm/day.
$J_{w[z]}$ — water flux density, cm^3 solution/(cm^2 soil, day).
$J_{s[z]}$ — solute flux density, meq solute/(cm^2 soil, day).
λ — dispersivity coefficient, cm.
L — length of active root zone.
LF — leaching fraction, (cm^3 drainage/day)/((cm^3 irrigation)/day).
θ_{im} — immobile volumetric water content, cm^3 immobile water/cm^3 soil.
θ_m — mobile volumetric water content, cm^3 mobile water/cm^3 soil.
θ_v — volumetric water content cm^3 water/cm^3 soil.
R — water uptake shape parameter, $\frac{S_w(L)}{S_w(O)}$.
$S_{w[z]}$ — water uptake function, cm^3 solution/(cm^3 soil, day).
t — time, days.
V — average pore water velocity, cm/day.
$V_{[z]}$ — average pore water velocity at a depth z, cm/day.
y — dimensionless depth.
z — depth, cm.